

1 Discussion Questions

- (a) What is the difference between Forward Checking and AC-3?

Forward checking and AC-3 both enforce arc consistency, but forward checking is more limited. Whenever a variable X is assigned, forward checking enforces arc consistency only for arcs that are pointing to X , which will reduce the domains of the neighboring variables in the constraint graph. Forward checking stops at this point, but AC-3 will continue to enforce arc consistency on neighboring arcs until there are no more variables whose domain can be reduced. As a result, FC ensures arc consistency of the assigned variable and its neighbors only, while AC-3 ensures arc consistency for the whole graph.

- (b) Why would one use the following heuristics for CSP?

- (i) Minimum Remaining Values (MRV)

MRV: “Which variable should we assign next?”

- Fail fast
- We have to assign all variables at some point, so we might as well do hard stuff first (allowing us to prune the search tree faster/realize we need to backtrack)

- (ii) Least Constraining Value (LCV)

LCV: “Which value should we try next?”

- We just want one solution.
- We don’t try all combinations of value, so we should try ones that are likely to lead to a solution.

2 CSP: Air Traffic Control

We have five planes: A, B, C, D, and E and two runways: international and domestic. We would like to schedule a time slot and runway for each aircraft to either land or take off. We have four time slots: 1, 2, 3, 4 for each runway, during which we can schedule a landing or take off of a plane. We must find an assignment that meets the following constraints:

- Plane B has lost an engine and must land in time slot 1.
- Plane D can only arrive at the airport to land during or after time slot 3.
- Plane A is running low on fuel but can last until at most time slot 2.
- Plane D must land before plane C takes off, because some passengers must transfer from D to C.
- No two aircrafts can reserve the same time slot for the same runway.

(a) Complete the formulation of this problem as a CSP in terms of variables, domains, and constraints (both unary and binary). Constraints should be expressed implicitly using mathematical or logical notation rather than with words. Make sure to specify variables, domains, and constraints.

Variables: A, B, C, D, E for each plane.

Domains: a tuple (*runway type*, *time slot*) for runway type $\in \{\text{international, domestic}\}$ and time slot $\in \{1, 2, 3, 4\}$.

Constraints:

$$B[1] = 1$$

$$D[1] \geq 3$$

$$A[1] \leq 2$$

$$D[1] < C[1]$$

$$A \neq B \neq C \neq D \neq E$$

Note here we use $B[1]$ to denote the second value of the tuple assigned to variable B, the time slot value, which is a number in $\{1, 2, 3, 4\}$.

For the following parts, we add the following two constraints:

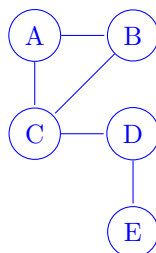
- Planes A, B, and C cater to international flights and can only use the international runway.
- Planes D and E cater to domestic flights and can only use the domestic runway.

(b) The addition of the two constraints above alters the CSP. Specifically, the domain does not need to include the runway type since this information is carried by the variable, and the binary constraints have changed. Determine the new domain and complete the constraint graph for this problem given the original constraints and the two added ones.

Variables: A, B, C, D, E for each plane.

Domain: $\{1, 2, 3, 4\}$

Constraint Graph:



Explanation of Constraints Graph: We can now encode the runway information into the identity of the variable, since each runway has more than enough time slots for the planes it serves. We represent the non-colliding time slot constraint as a binary constraint between the planes that use the same runways.

(c) What are the domains of the variables after enforcing arc consistency? Begin by enforcing unary constraints. (Cross out values that are no longer in the domain.)

Enforcing arc consistency with AC-3, we have the following domain as a result:

A	1	2	3	4
B	1	2	3	4
C	1	2	3	4
D	1	2	3	4
E	1	2	3	4

(explanation of process below)

Enforcing unary constraints (in an arbitrary order) first,

1. We cross out 2, 3, 4 from B's domain, adding arcs $A \rightarrow B$ and $C \rightarrow B$ to the queue.
2. We cross out 3, 4 from A's domain, adding arcs $B \rightarrow A$ and $C \rightarrow A$ to the queue.
3. We cross out 1, 2 from D's domain, adding arcs $C \rightarrow D$ and $E \rightarrow D$ to the queue.

Enforcing $A \rightarrow B$, we cross out 1 from A's domain; Arcs $B \rightarrow A$ and $C \rightarrow A$ are already on the queue.

Enforcing $C \rightarrow B$, we cross out 1 from C's domain; add arcs $A \rightarrow C$, $B \rightarrow C$, and $D \rightarrow C$ to the queue.

Enforcing $B \rightarrow A$, no domain changes are necessary (all values remaining in B's domain have a consistent corresponding value in A's domain); no arcs are added.

Enforcing $C \rightarrow A$, we cross out 2 from C's domain; Arcs $A \rightarrow C$, $B \rightarrow C$, and $D \rightarrow C$ are already on the queue.

Enforcing $C \rightarrow D$, we cross out 3 from C's domain; Arcs $A \rightarrow C$, $B \rightarrow C$, and $D \rightarrow C$ are already on the queue.

Enforcing $E \rightarrow D$, no domain changes are necessary.

Enforcing $A \rightarrow C$, no domain changes are necessary.

Enforcing $B \rightarrow C$, no domain changes are necessary.

Enforcing $D \rightarrow C$, we cross out 4 from D's domain (there is no c in C's domain such that $c > 4$); add arcs $C \rightarrow D$ and $E \rightarrow D$ to the queue.

Enforcing $C \rightarrow D$, no domain changes are necessary.

Enforcing $E \rightarrow D$, we cross out 3 from E's domain; add arc $D \rightarrow E$ to the queue.

Enforcing $D \rightarrow E$, no domain changes are necessary.

(phew!)

Note: For a general binary CSP, to enforce arc consistency before assigning any variables, you should add all arcs to the initial queue. For this problem, it can be easily seen that if there are no unary constraints, all the arcs will be consistent before any variable is assigned a value. As a result, we can start with the unary constraints and add arcs only for the related variables after enforcing the unary constraints.

(d) Arc-consistency can be rather expensive to enforce, and we believe that we can obtain faster solutions using only forward-checking on our variable assignments. Using the Minimum Remaining Values heuristic, perform backtracking search on the graph, breaking ties by picking lower values and characters first. List the (variable, assignment) pairs in the order they occur (including the assignments that are reverted upon reaching a dead end). Enforce unary constraints before starting the search.

List of (variable, assignment) pairs:

(You don't have to use this table)

A		1	2	3	4
B		1	2	3	4
C		1	2	3	4
D		1	2	3	4
E		1	2	3	4

Answer: (B, 1), (A, 2), (C, 3), (C, 4), (D, 3), (E, 1)

3 Schedule classes to professors

You are in charge of finalizing the schedule for the AI classes being offered for this semester. Your task is to assign classes to professors. There are 5 classes and 3 professors who are available for teaching these classes. You are constrained by the fact that each professor can only teach one class at a time and each professor is only qualified to teach a subset of the classes being offered.

The classes are:

1. Class 1 - Automated planning: scheduled for 8:00-9:00am
2. Class 2 - Intro to Artificial Intelligence: scheduled for 8:30-9:30am
3. Class 3 - Natural Language Processing: scheduled for 9:00-10:00am
4. Class 4 - Computer Vision: scheduled for 9:00-10:00am
5. Class 5 - Machine Learning: scheduled for 10:30-11:30am

The professors are:

1. Professor A, who is qualified to teach Classes 1, 2, and 5;
 2. Professor B, who is qualified to teach Classes 3, 4, and 5;
 3. Professor C, who is qualified to teach Classes 1, 3, and 4.
1. Formulate this problem as a CSP problem in which there is one variable per class, stating the domains (after enforcing unary constraints), and binary constraints. The solution of the problem shall determine the assignment of classes to professors.

Constraints should be specified formally (i.e., using math or logic expressions) and precisely, but may be implicit rather than explicit.

Solution:

Variables are the classes and Domains are the professors qualified for each class:

$C_1: \{A, C\}$,

$C_2: \{A\}$,

$C_3: \{B, C\}$

$C_4: \{B, C\}$

$C_5: \{A, B\}$

Constraints are based on the time / overlapping of the courses and we conveniently express them as binary constraints.

Binary Constraints:

$C_1 \neq C_2$

$C_2 \neq C_3$

$C_2 \neq C_4$

$C_3 \neq C_4$

2. Draw the constraint graph associated with the CSP model.

Solution:

